

Eight years of autonomic monitoring: An N-of-1 longitudinal study of wearable-derived HRV anomalies and self-reported mood logs

Jeff Brozena¹ Saeed Abdullah¹ Lan Kong² Erika F.H. Saunders² Guodong Liu²
¹Penn State University ²Penn State University College of Medicine

Introduction

Self-tracking can benefit individuals with bipolar disorder. Technology-enabled self-tracking has been shown to improve self-reflection and feelings of agency in this population [1].

Given the significant associations between sleep and bipolar disorder [2], passive collection of sleep data can play a role in illness self-management. For example, measures of circadian instability and nocturnal HRV anomaly [3] can be derived from data collected by increasingly-accurate sensors available in consumer devices such as the Oura Ring [4].

This exploratory study (individual with bipolar disorder, N-of-1) demonstrates the feasibility of collecting longitudinal illness-related sleep and mode data over nearly a decade. We assess temporal dynamics across mood states and nightly HRV anomalies contained within the dataset.

Methods

Physiological data were collected via Oura Ring (gen. 2 and 3) and daily symptoms via the iOS app eMoods Tracker. HRV (SDNN) and mood were aligned to a shared daily timeline. We derived a lagged 90-day rolling z-score for modeling purposes.

Data were fit using vector autoregression (VAR) models on a stationarity-tested series. We performed pairwise Granger causality tests across all lags with FDR-BH correction (alpha=0.05).

Results are shown as impulse response functions for mood “shocks” and as a consensus network of significant directed temporal relationships.

Dataset Contents

Data structure	Description
Wearable-derived sleep data	Feb 2019–Sept 2026 4,077 recorded nights 98.7% complete
Self-report mood logs	Aug 2019–Sept 2026 712 mood entries Ordinal scale (0–3) Elevated, depressed, anxious, irritable

Prevalence of reported mood states

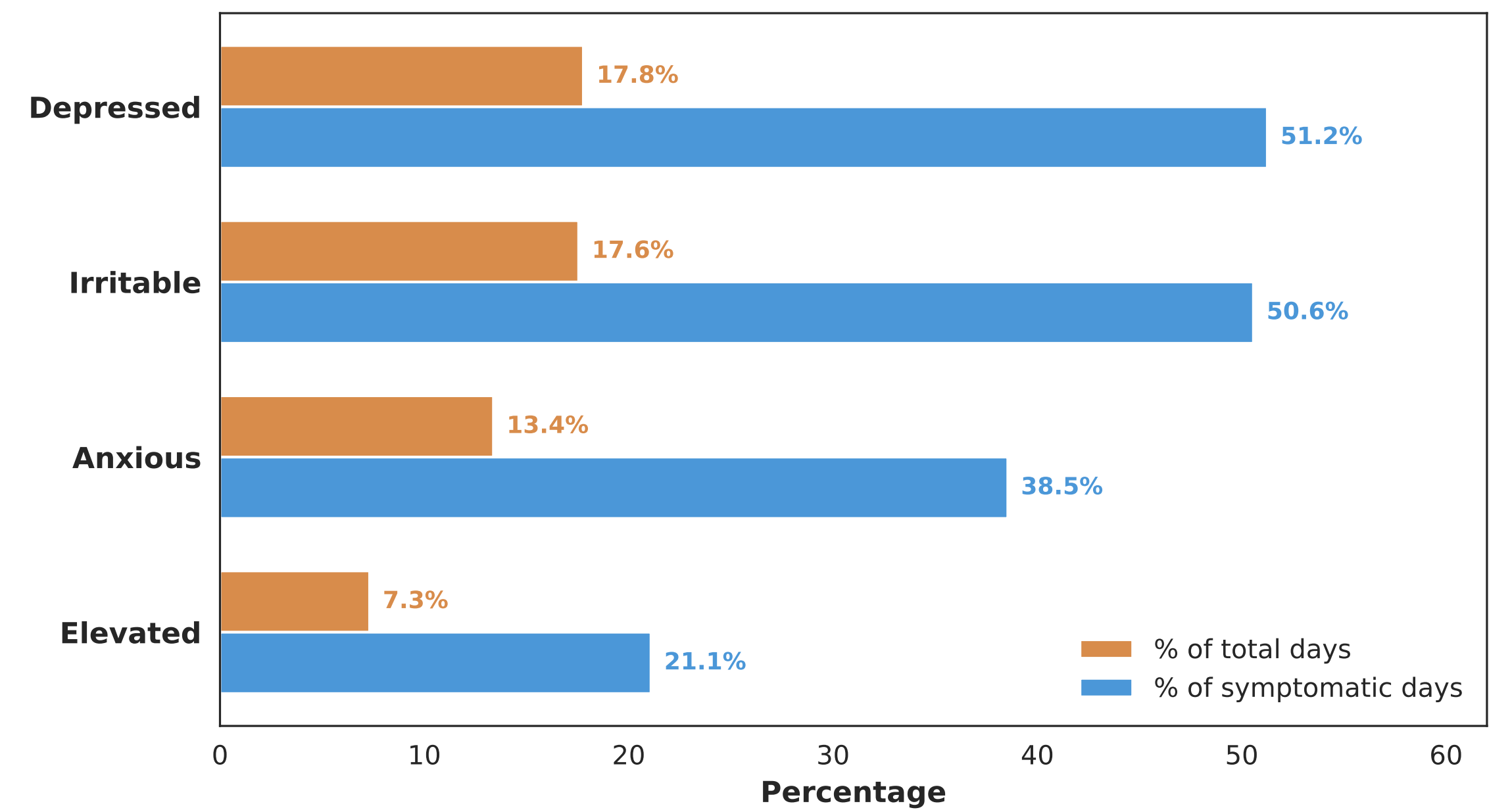


Figure 1. Prevalence of each self-reported mood state as percentage of all logged days (N=1,127) and of symptomatic days (n=441). Depressed and irritable states were reported most frequently. Depressed states comprised over half (51.2%) of symptomatic days.

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Consensus network of significant directed temporal relationships

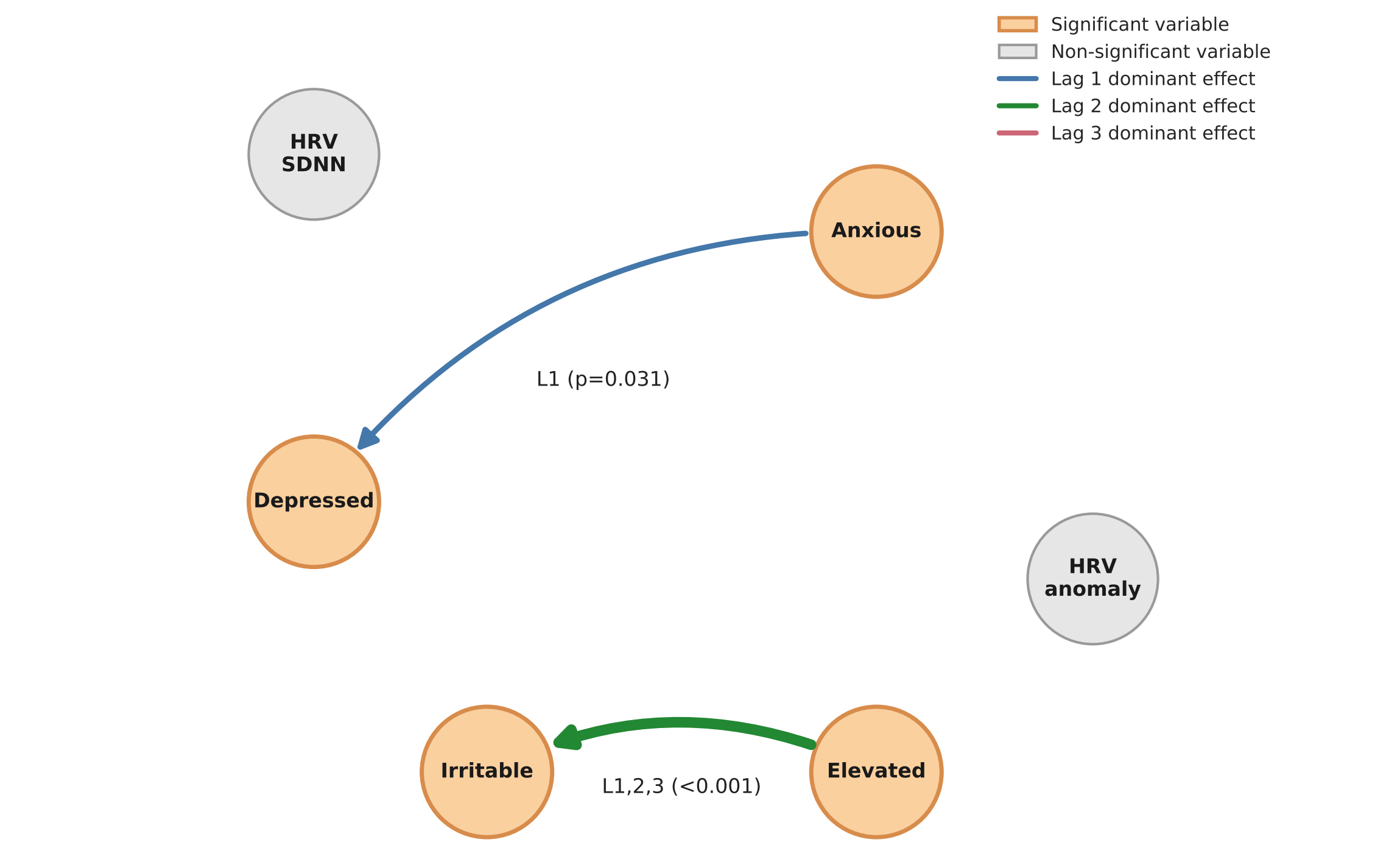


Figure 2. Directed graph of the consensus Granger-causality network. Each lag represents one day. Arrows show the direction of lagged temporal predictability retained after FDR-BD correction (alpha = 0.05). Edges reflect temporal predictability rather than a causal relationship.

Lagging temporal effects between mood states

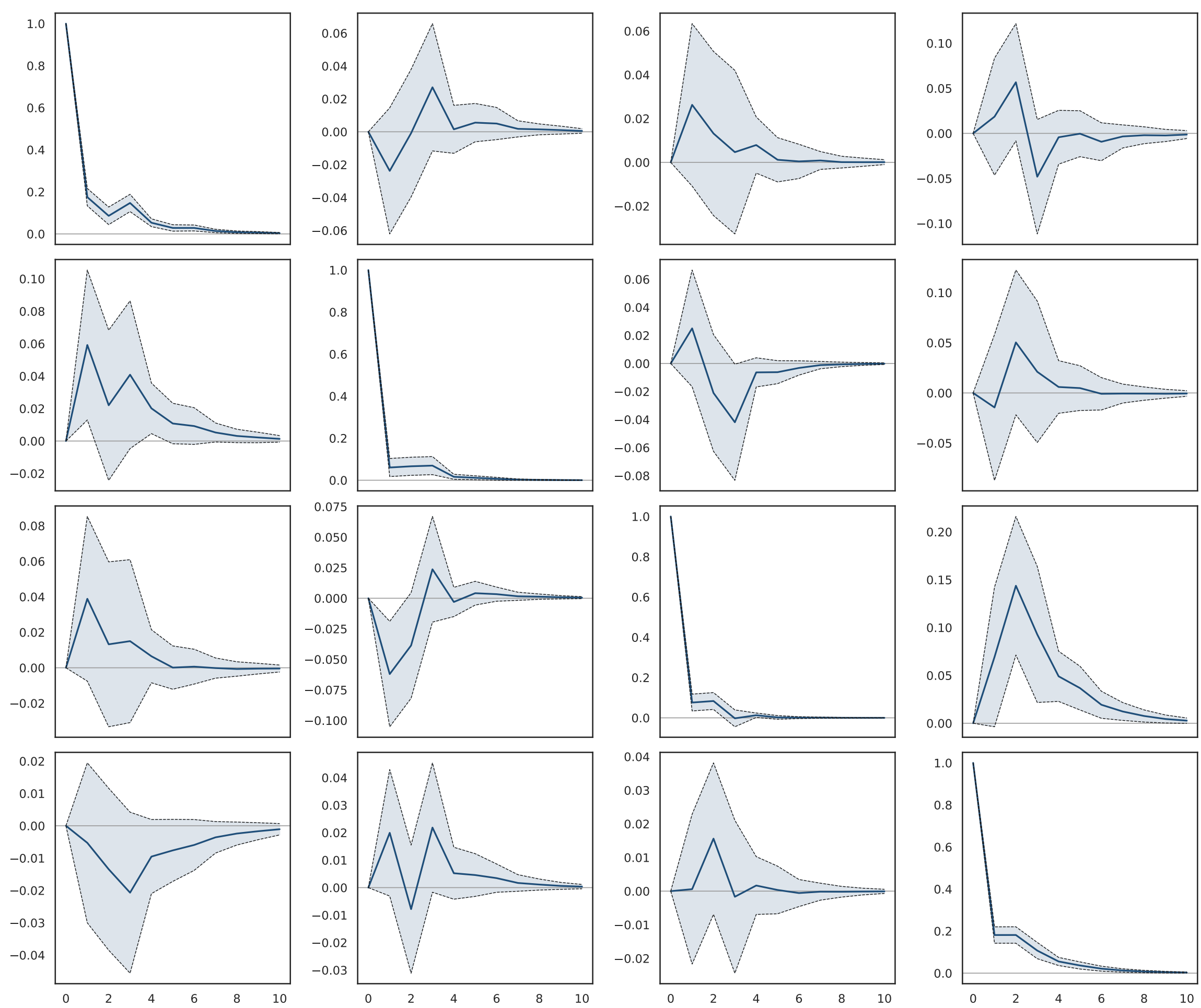


Figure 3. Impulse response functions (IRFs) showing the estimated response of each mood state over a 10-day horizon to a one-unit shock in each mood state. Diagonal panels capture each state’s self-persistence. Off-diagonal panels indicate between-mood dynamics across lags. Shaded bands indicate 95% confidence intervals.

Significant dynamics surviving FDR correction were exclusively mood-to-mood. Anxious moods predicted next-day depressed moods (lag-1, $p = 0.031$) while elevated moods predicted irritability across all three lags ($p < 0.001$). Dynamics involving HRV anomaly did not survive FDR correction in our model.

References

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